

## 20. Linear Predictors (part 3: logistic regression)

Monday, February 19, 2024 5:25 PM

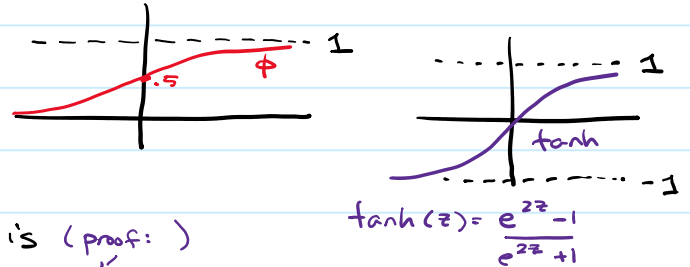
It's not really **regression**, it's **classification** where instead of outputting a label 0 or 1 (or  $\pm 1$ ), we output a **probability** reflecting our confidence in a label

$\mathcal{H} = \phi \circ \mathcal{L}_d$  with **"sigmoid"**

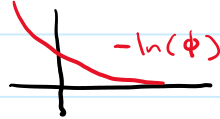
$$\phi(z) := \frac{1}{1+e^{-z}} = \frac{e^z}{1+e^z}$$

often denoted  $\sigma$

$$= \frac{1}{2} (\tanh(z) + 1)$$



- $\phi$  isn't convex but  $-\ln(\phi(z))$  is (proof:)



$$f(z) = -\ln(\phi(z)) = \ln\left(\frac{1}{\phi(z)}\right) = \ln(1+e^{-z})$$

So  $f'(z) = \frac{-e^{-z}}{1+e^{-z}}$  and  $f''(z) = \frac{e^{-z}(1+e^{-z}) + e^{-z}(-e^{-z})}{(1+e^{-z})^2}$

$$= \frac{e^{-z}}{(1+e^{-z})^2} > 0 \text{ So convex}$$

$\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}$

- $\phi^{-1}(p) = \ln\left(\frac{p}{1-p}\right) =: \text{logit}(p)$  if  $p \in (0,1)$

- $2\phi(z) - 1 \approx \text{sign}(z)$  if  $|z| \gg 1$

$\tanh(z) \approx \text{sign}(z)$  when  $|z| \gg 1$

- $\phi(z) = 1 - \phi(-z)$  symmetry

Motivation / justification for loss function

Switching to  $\sigma$  notation ( $\sigma = \phi$ )

Assume  $\mathcal{D}_{y|x}$  is given by  $y = \begin{cases} +1 & \text{w.p. } \phi(\langle \omega^*, x \rangle) \\ -1 & \text{w.p. } 1 - \phi(\langle \omega^*, x \rangle) \end{cases}$

or concisely,  $\mathbb{P}[Y=y | \omega^*, x] = \phi(y \cdot \langle \omega^*, x \rangle)$  since  $\phi(z) = 1 - \phi(-z)$

an example of a Generalized Linear Model (GLM)

$$\mathbb{P}[Y=y | x] = g^{-1}(\langle \omega^*, x \rangle)$$

["General" Linear Model just means several columns and Frobenius norm]

then if we have  $m$  iid observations,

the likelihood is  $\mathbb{P}[(y_i)_{i=1}^m | \omega, (x_i)_{i=1}^m] = \prod_{i=1}^m \mathbb{P}[y_i | \omega, x_i]$

and the **maximum likelihood estimator (MLE)** maximizes this likelihood (or minimize negative log (likelihood))

$$\begin{aligned} \omega_{MLE} &\in \underset{\omega}{\text{argmin}} - \sum_{i=1}^m \ln(\sigma(y_i \langle \omega, x_i \rangle)) \\ &= \underset{\omega}{\text{argmin}} + \sum_{i=1}^m \ln(1 + \exp(-y_i \langle \omega, x_i \rangle)) \end{aligned}$$

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Monday, February 26, 2024 9:38 AM

So if we choose our loss function  $l(\hat{y}, y) = \log(1 + e^{-y\hat{y}})$

then ERM is just MLE

So impossible to achieve 0-loss

(and logistic regression is well-understood; see ch. 24 for more on MLE)

... and is a convex problem!

### Random numerical issues

Compute  $f(a) = \ln(1 + e^a)$  for  $a = 400 \cdot \ln(10)$  (so  $e^a = 10^{400}$ )

it should be  $f(a) \approx a$  since  $1 + e^a \approx e^a$  and  $\ln(e^a) = a$

but numerically, the intermediate term  $e^a$  produces overflow

Instead:

$$\ln(1 + e^a) = \ln(e^a(e^{-a} + 1)) = a + \ln(1 + e^{-a})$$

underflows

and calculate

$$\ln(1 + e^{-a}) = \log_{10}(e^{-a})$$

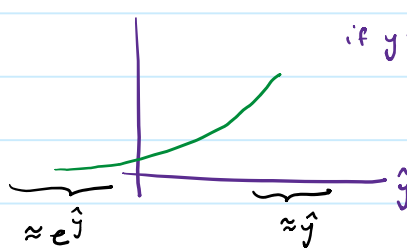
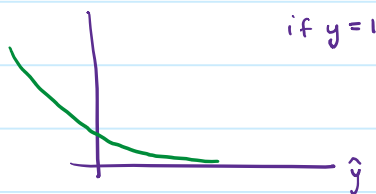
which uses Taylor series

$$\log(1+x) = x - x^2/2 + x^3/3 - \dots$$

Similarly: `numpy.logaddexp`

for  $\log(e^a + e^b)$

plots of  $l(\hat{y}, y)$



if  $y=-1 \rightarrow$  if  $\hat{y} \ll 0$ ,  $e^{\hat{y}} \approx 0$

$$\log(1+x) = x - x^2/2 + x^3/3 - \dots$$

$$\log(1 + e^{\hat{y}}) \approx e^{\hat{y}}$$

$\rightarrow$  if  $\hat{y} \gg 1$ ,  $e^{\hat{y}} \gg 1$

$$\text{so } \log(1 + e^{\hat{y}}) \approx \log(e^{\hat{y}}) = \hat{y}$$